Salt Marsh Vegetation at North Campus Open Space

Dominant species, native/non-native cover, and restoration trajectories

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Introduction

North Campus Open Space (NCOS) is a restored coastal wetland system in Santa Barbara, California, created from the former Ocean Meadows Golf Course and designed to support a mosaic of wetland and upland habitats, including salt marsh, brackish marsh, seasonal wetlands, and transitional habitats (Cheadle Center for Biodiversity and Ecological Restoration and Environmental Science Associates 2016; Rickard 2024). Because restored wetlands develop over time, vegetation monitoring provides a useful way to evaluate whether habitat structure is moving toward intended ecological conditions. In salt marshes especially, plant cover and species identity can reveal more than whether vegetation is present; they can show which species are shaping habitat identity, whether native halophytes are establishing, and how non-native species contribute to community composition.

Salt marsh vegetation is especially useful for understanding restoration trajectory because these plant communities are shaped by strong environmental filters, including salinity, tidal flooding, saturated soils, elevation, and seasonal changes in water availability. In southern California salt marshes, native salt-tolerant species such as pickleweed (Salicornia pacifica) and saltgrass (Distichlis spicata) often help define the structure and identity of coastal marsh habitat (Ferren 1985; North Campus Open Space 2025). Because these species are adapted to stressful coastal wetland conditions, changes in their mean percent cover can help indicate whether a restored salt marsh is developing toward a recognizable native marsh community.

This question is particularly relevant at NCOS because monitoring reports evaluate restoration progress across multiple habitat types, including vegetation cover, native biodiversity, and relative native cover (Rickard 2024). However, broad habitat-level summaries can sometimes obscure the species-level patterns that determine what kind of plant community is forming. A marsh with high total vegetation cover may still be ecologically different depending on whether that cover is dominated by native salt marsh species or by non-native plants. For this reason, examining dominant species and native/non-native cover together can provide a more detailed view of restoration progress than total vegetation cover alone.

Previous work on California wetland restoration shows that vegetation establishment can be slow, variable, and strongly shaped by local site conditions such as elevation, salinity, soil moisture, planting, disturbance, and adaptive management (Beheshti et al. 2023). These factors matter because shifts in plant cover are not just visual patterns in a graph. They may reflect ecological processes such as native plant establishment, non-native plant expansion, hydrologic change, disturbance, or interannual variability in rainfall and growing conditions. Interpreting vegetation cover as part of a restoration trajectory therefore requires attention to both species composition and change through time.

This study focuses on salt marsh and closely related brackish marsh vegetation at NCOS. We ask three main questions: Which dominant plant species characterize salt marsh and brackish marsh vegetation at NCOS? How has mean percent cover of dominant salt marsh species changed across survey years? How has native and non-native vegetation cover changed across survey years? Together, these questions allow us to connect vegetation monitoring data to broader questions about restoration progress, habitat identity, and community development in a restored coastal wetland.

Methods

Study system and datasets

We used vegetation monitoring data from NCOS to examine plant community composition in salt marsh and brackish marsh habitat. The main dataset, veg.csv, contains vegetation survey observations, including plant species identity, percent cover, cover category, habitat type, transect, site, date, and year. The metadata file, vp_veg_metadata.csv, provides supporting information about vegetation transects, including the number of quadrats sampled within each transect-year.

The dataset covers vegetation surveys from 2021 through 2025 in the filtered salt marsh and brackish marsh habitat categories. To make survey years more comparable, we focused on August and September surveys, when late-summer and early-fall vegetation cover is expected to be relatively high. We filtered the dataset to include only native and non-native plant cover categories, excluding bare ground, thatch, and other non-plant cover types.

Data cleaning and summary approach

We summarized vegetation cover as mean percent cover by year, transect, species, and cover category. Because sampling effort differed among transects and years, we used the metadata file to account for the number of quadrats sampled in each transect-year. We also used complete() to make implicit absences explicit as zero percent cover, so that species not recorded in a transect-year were treated as absent rather than missing.

Dominant species were identified using the 90th percentile of mean percent cover across all plant species in the filtered dataset. Species at or above this threshold were classified as dominant because their average cover exceeded that of most species recorded in the salt marsh/brackish marsh data. We then summarized dominant species cover by survey year and visualized these patterns with a stacked bar plot.

To compare native and non-native vegetation, we summarized transect-level native and non-native cover by year and then calculated yearly means and standard errors. We also summarized total vegetation cover by year to provide broader context for changes in vegetation abundance through time. These analyses are descriptive and are intended to identify patterns in community composition and cover through time.

Results

Dominant species identity

Species differed strongly in mean percent cover across the filtered salt marsh and brackish marsh dataset. Salicornia pacifica had the highest mean cover overall, followed by Distichlis spicata and Atriplex lentiformis. Several non-native species, including Raphanus sativus and Festuca perennis, also appeared among the higher-cover species, showing that non-native plants contributed meaningfully to marsh vegetation composition even when native species dominated overall.

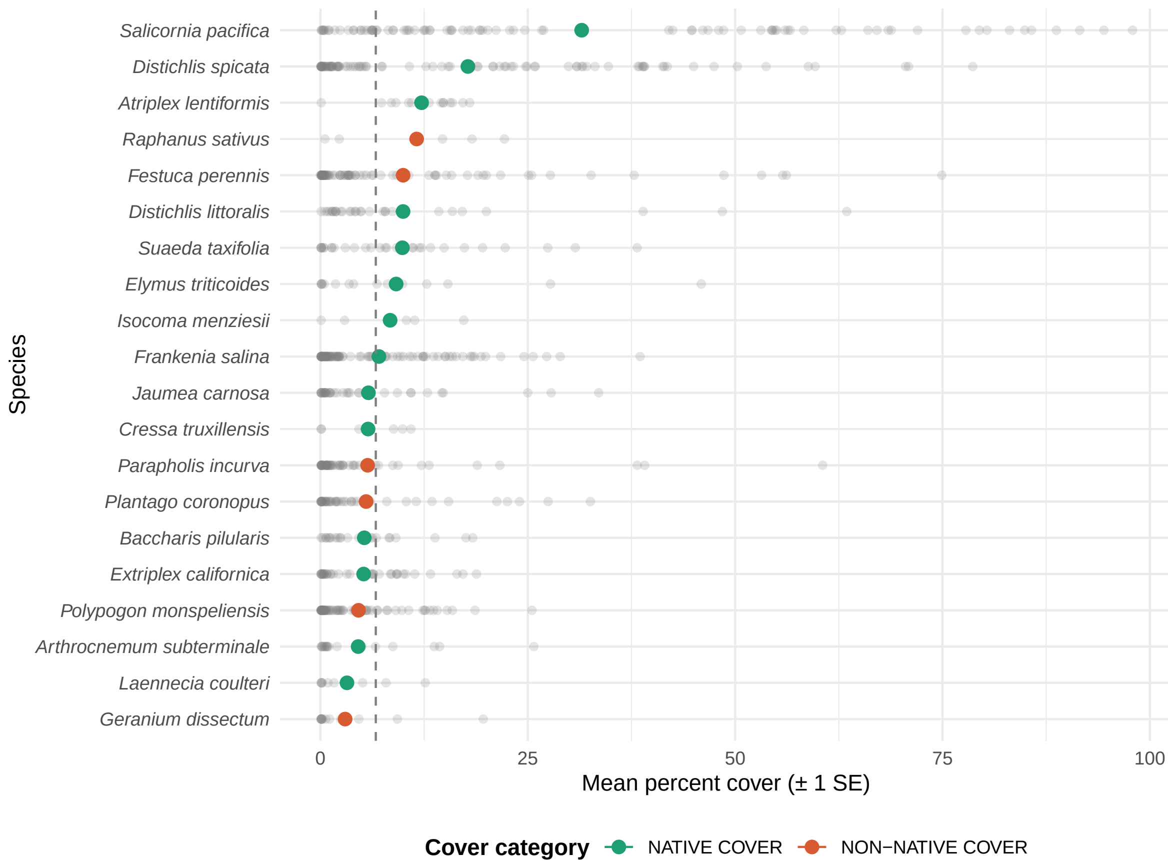

Figure 1: Mean percent cover (± 1 SE) of the top 20 plant species recorded in filtered salt marsh/brackish marsh habitat at NCOS. Species are ranked by mean cover across transects and survey years. Green indicates native species and coral indicates non-native species. The dashed line marks the 90th percentile cutoff used to identify dominant species.

Dominant species cover through time

Dominant species cover varied across survey years. The summed mean cover of dominant species was highest around 2023 and lower in 2025, suggesting that dominant vegetation cover may fluctuate across years rather than increasing steadily. Native salt marsh species, especially Salicornia pacifica and Distichlis spicata, were major contributors across the survey period. However, non-native dominant species also contributed to cover in some years, especially Raphanus sativus and Festuca perennis.

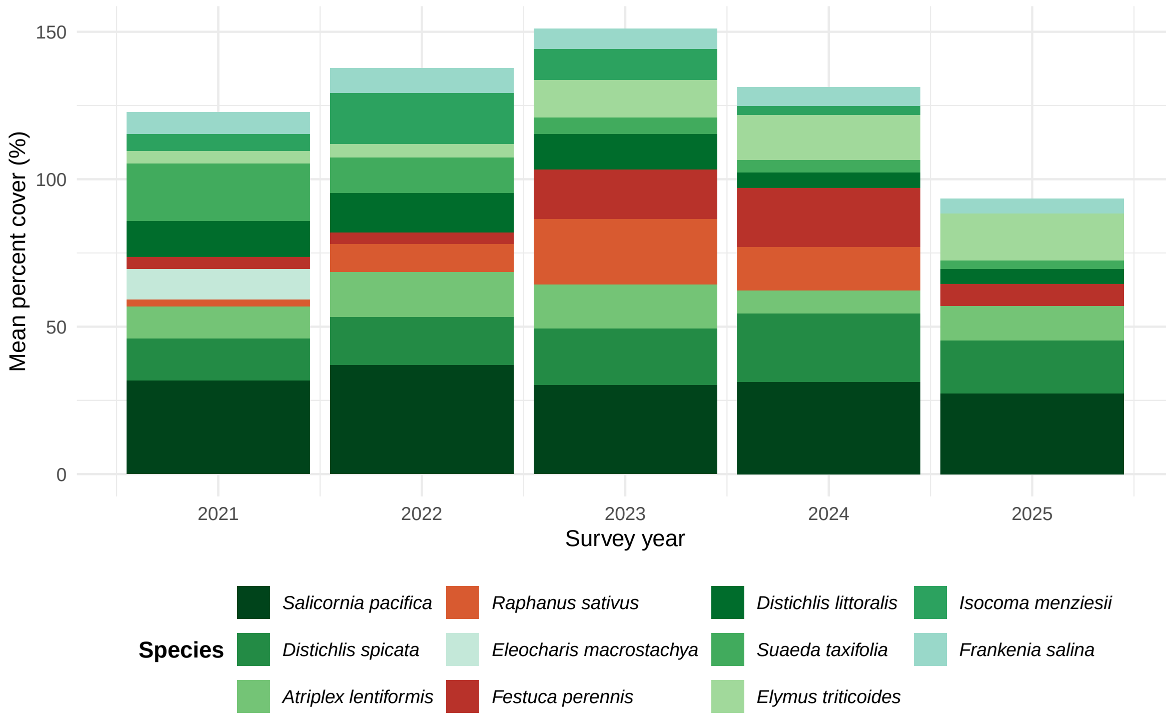

Figure 2: Mean percent cover of dominant salt marsh/brackish marsh plant species at NCOS across survey years. Dominant species were identified as species at or above the 90th percentile of mean percent cover across all survey years. Bar height reflects total mean cover of the dominant species and varies across years. Green shades indicate native species and coral/red shades indicate non-native species.

Native and non-native vegetation cover

Native vegetation cover remained higher than non-native vegetation cover across all survey years. Native cover increased from 2021 to 2022 and remained relatively high through 2024 before declining in 2025. Non-native cover was lower overall but increased from 2022 to

2024 before also declining in 2025. These patterns suggest that native vegetation was the stronger component of marsh cover, while non-native vegetation still contributed noticeably to community composition.

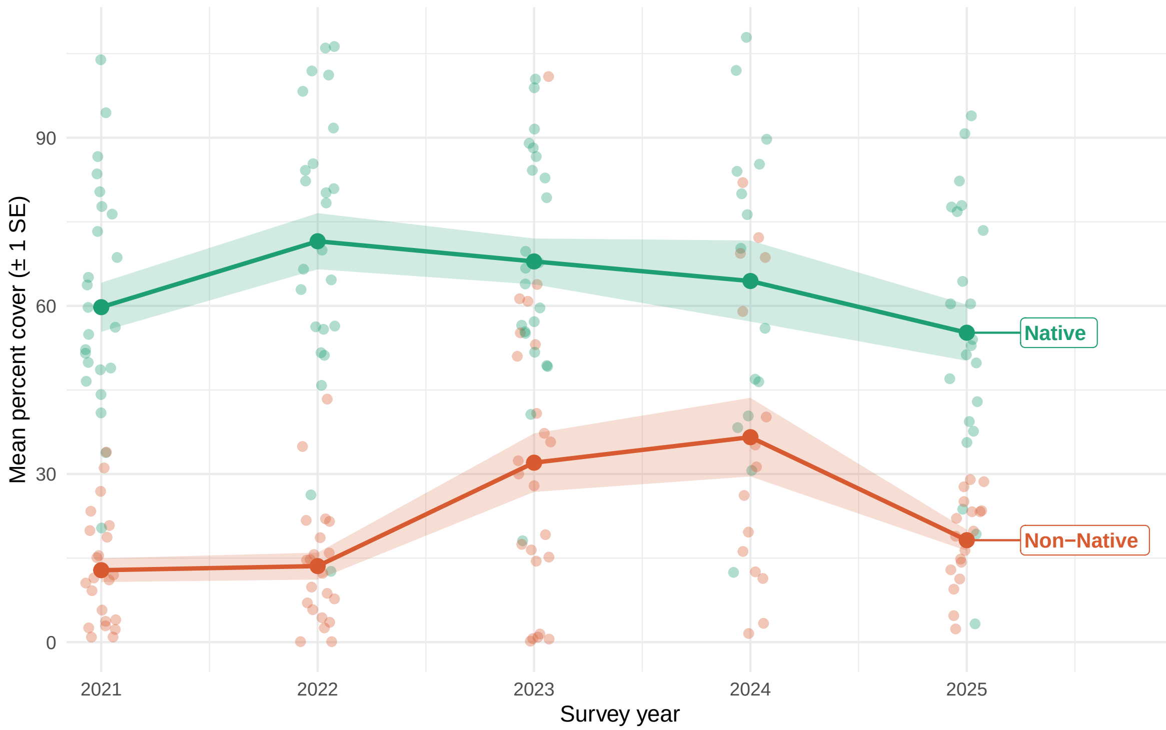

Figure 3: Mean percent cover (± 1 SE) of native and non-native vegetation in salt marsh/brackish marsh habitat at NCOS across survey years. Faint points show transect-level observations; lines connect yearly means; shaded bands represent ± 1 standard error. Native cover is shown in green and non-native cover in coral.

Total vegetation cover

Total vegetation cover, combining native and non-native plants, increased from 2021 to 2024 and then declined in 2025. This broader pattern provides context for the species-level and native/non-native patterns above: changes in relative composition occurred alongside changes in total cover, suggesting that interannual variability may influence both overall vegetation abundance and the species contributing to that cover.

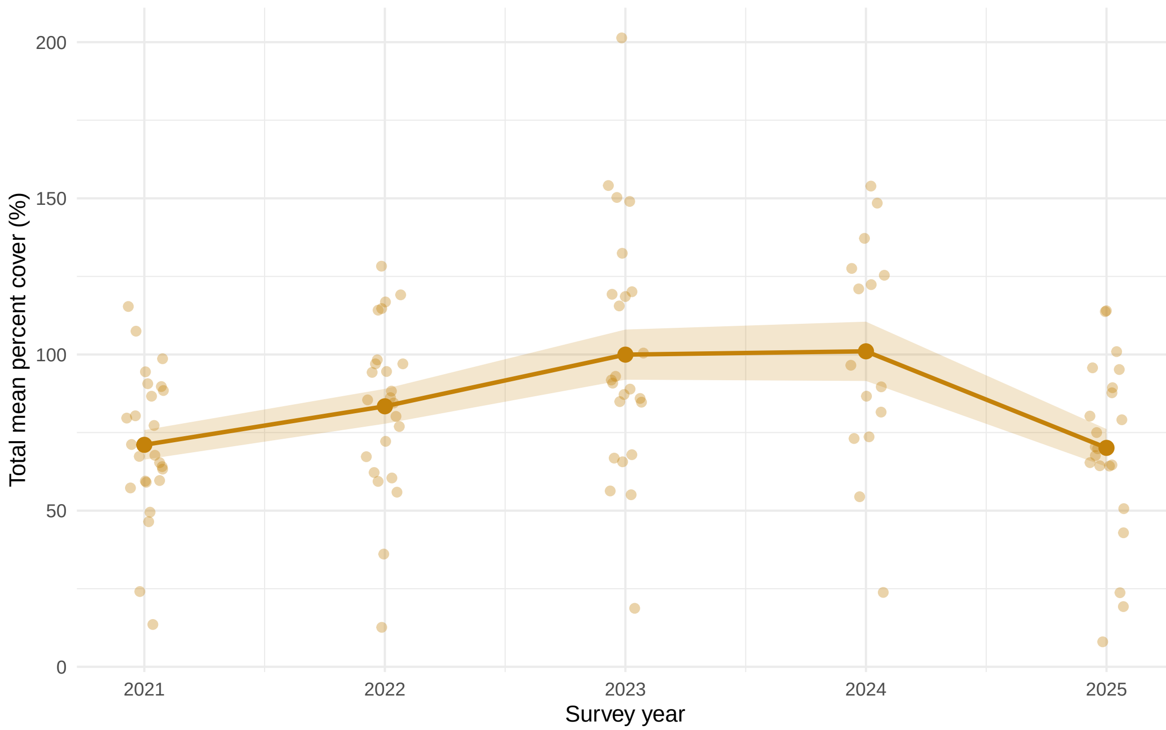

Figure 4: Total mean percent vegetation cover (native + non-native combined) in salt marsh/brackish marsh habitat at NCOS across survey years. Faint points show transect-level observations; the line connects yearly means; the shaded band represents ± 1 standard error around the mean.

Discussion

The vegetation patterns found in this project align with the NCOS monitoring report and its description of a restored salt marsh habitat (Rickard 2024). Along with noting that all salt marsh habitats showed only slight changes in native and non-native cover between Year 6 and Year 7, the monitoring report identifies *Salicornia pacifica* and *Distichlis spicata* as the most frequently occurring native species in quadrat monitoring which our figures also showed as dominant. Furthermore, the report documents that salt marsh habitats met total cover and relative native cover success criteria (70% and 71% relative native cover in 2024, respectively), which agrees with our finding that native vegetation consistently exceed non-native cover across survey years. Moreover, the report also notes that absolute native cover criteria were not fully met in 2024 due in part to non-native annual grasses, a pattern our figures reflect in the non-trivial non-native cover, particularly from *Festuca perennis* and *Raphanus sativus*. These patterns together suggest that the NCOS restored salt marsh is establishing toward a native halophyte community while still facing persistent non-native cover; a trajectory consistent with restored southern California salt marshes more broadly (Callaway and Zedler 2004; Beheshti et al. 2023).

The challenges in collecting and summarizing the field monitoring data used in this project may influence the interpretation of the vegetation patterns found. The NCOS field data collection can be challenging as percent cover is estimated visually within quadrats and it is subject to observer bias and variability. Trained surveyors may disagree on values for species with overlapping canopies or variable growth forms like *Salicornia pacifica* (Thomsen et al. 2021). Furthermore, surveys focused on August and September miss earlier season growth dynamics and the effects on winter flooding on species composition, which have been shown to strongly influence halophyte cover in California marshes (Allison 1992). The decline in both native and total vegetation cover in 2025 is difficult to interpret without additional context, but the Year 7 monitoring report notes that two years of elevated precipitation and high slough water levels reduced lower-elevation salt marsh vegetation in some transects. This suggests that interannual hydrological variation that was not captured directly in the vegetation dataset likely plays a role in the cover fluctuations we observe. Also, variables such as soil salinity, elevation, and inundation frequency would provide important context for the species-level patterns in our figures (Beheshti et al. 2023; Sanderson, Foin, and Ustin 2001).

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